

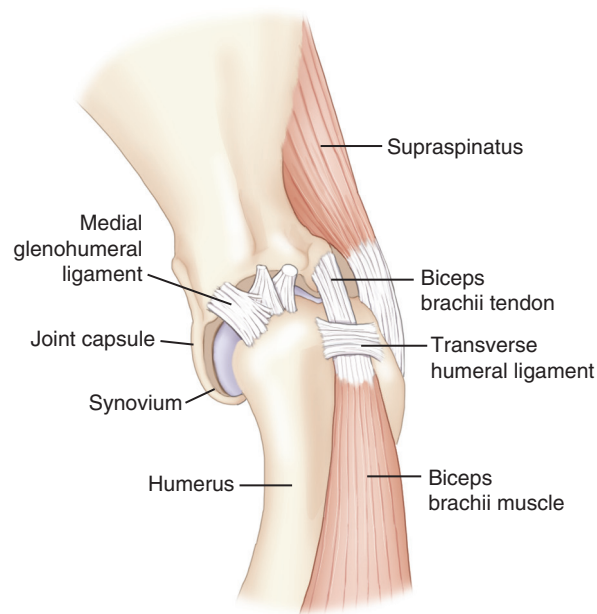


Figure 33-11 Normal canine shoulder anatomy (relevant to MSI).

tendon, supraspinatus tendon, labrum, and less commonly the biceps tendon (shown in [Figures 33-12](#) and [33-13](#)). Depending on the severity and chronicity of the condition, the cartilage may also be affected.

Cause

Currently, the exact cause of MSS in dogs is unknown, although it is suspected to be related to chronic repetitive activity or overuse rather than trauma. Overuse of the shoulder support structures leads to degeneration of the tissues, lowering the tensile strength of the tissues and predisposing them to fraying, disruption, and eventually complete breakdown.^{1,20} Sporting athletes that participate in activities such as agility undergo extreme stresses on their muscles, ligaments, and tendons. Repetitive activities such as jump-turn combinations and weave poles performed regularly during practice and at weekend trials place the shoulder near its end range of abduction (as shown in [Figure 33-14](#)), and stress the soft tissues of the medial shoulder complex. Additionally, events such as slipping on wet surfaces, mishaps on the dog walk, teeter,

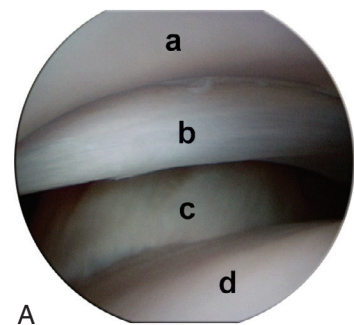
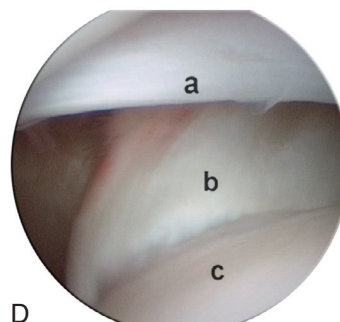
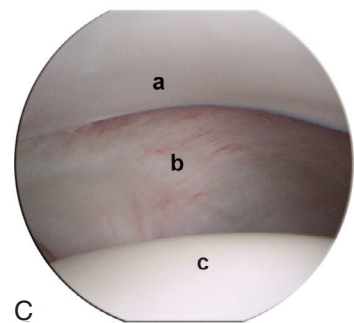
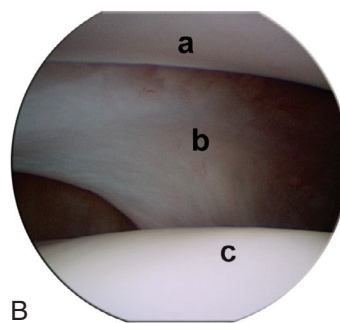


Figure 33-12 **A**, Arthroscopic view of the cranio-medial compartment of the shoulder joint. Glenoid (a), cranial arm of the medial collateral ligament (MCL) (b), subscapularis tendon (c), and humeral head (d). **B**, Arthroscopic view of the centromedial compartment of the shoulder joint. Glenoid (a), centromedial area of the MCL (b), and humeral head (c). **C**, Arthroscopic view of the caudomedial compartment of the shoulder joint. Glenoid (a), caudomedial area of the MCL (b), and humeral head (c). **D**, Arthroscopic view of the cranial compartment of the shoulder joint. Cranial arm of the MCL (a), insertion of the subscapularis tendon (b), and humeral head (c). (From Beale, B.; Hulse, D.; Schulz, K.; Whitney, W.: *Small Animal Arthroscopy*, 2003, Saunders.)



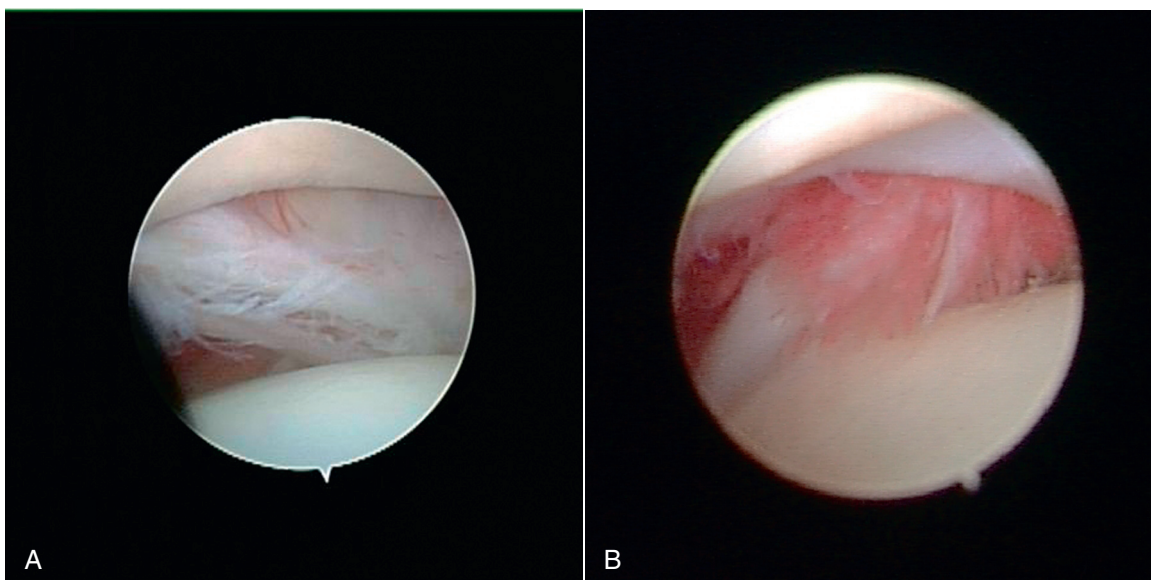


Figure 33-13 Abnormal arthroscopic images of an agility dog with MSI.



Figure 33-14 A shoulder in the end range of abduction.

or A-frame may also contribute to the trauma inflicted on the shoulder while participating in agility. Over time there may be a cumulative effect of the microtrauma occurring to the ligaments, tendons, and joint capsule, leading to a decrease in performance.

Diagnosis

History and Clinical Signs. Presenting complaints and clinical signs in dogs with MSS may vary from subtle, such as missing cues or refusing tight turns during performance, to an overt intermittent unilateral lameness. It is not uncommon for the history to include identification of restrictions, spasm, and trigger points, or mild atrophy in the affected shoulder by a therapist (veterinary rehabilitation therapist, veterinary massage therapist, or veterinary chiropractor) during routine sessions. Dogs with chronic conditions typically have a history of poor response to rest and NSAIDs, and increased lameness after exercise and heavy activity.

Orthopedic Examination. Findings from gait evaluation may range from a mildly shortened stride in the affected forelimb at a walk and a trot to significant weight-bearing lameness. Depending on chronicity, atrophy may

be noted in the affected shoulder muscle(s) on physical examination. Forelimb circumference may be decreased in the affected forelimb when compared with the contralateral unaffected forelimb. Dogs with MSS typically have restriction and decreased range of motion in shoulder extension. The most consistent physical exam findings in dogs with MSS are shoulder spasm and discomfort on abduction of the affected shoulder. If a concurrent supraspinatus tendinopathy is present, pain may be noted when placing the shoulder into flexion (direct stretch of the supraspinatus) or on direct palpation of its point of insertion.

Because preoperative diagnosis of MSS is critical for clinical decision making, a palpation technique has been developed known as the *shoulder abduction test*.⁴⁸ Although this test was originally described in dogs under sedation in lateral recumbency, it is noted by the author that this test may be performed with similar consistent results on dogs awake and standing. To appropriately engage the components of the craniomedial shoulder, the elbow and shoulder must be placed in full extension with concurrent abduction of the forelimb. The scapula at the level of the acromion process must be stabilized with the evaluator's opposite hand to achieve a passive stretch of the craniomedial shoulder components. A goniometer is held such that one arm of the goniometer runs parallel to the spine of the scapula so that the center of rotation of the goniometer is centered over the shoulder joint and the free arm of the goniometer extends over the axis of the humerus in contact with the skin. With the elbow and shoulder held in extension, the forelimb is abducted to its physiologic limit and the measurement obtained as shown in Figure 33-15. In severe cases, which clinically appear to be much less common, a slight “thud” or subluxation may be felt when abducting the shoulder. The procedure should then be performed on the contralateral shoulder for comparison. Using this



Figure 33-15 Abduction test.

palpation technique, it has been reported that a normal mean shoulder abduction angle is approximately 30 degrees.⁴⁸ Although this report revealed that dogs with medial shoulder instability had abduction angles significantly larger than 30 degrees,⁴⁸ a more recent study showed an inconsistent correlation between abduction angles and medial compartment pathology identified arthroscopically.⁴⁹ This is also the author's clinical observation. Dogs with MSS rarely show evidence of "instability," but rather show signs of dysfunction and pain associated with cranio-medial pathology, including varying degrees of subscapular tendinopathy; medial glenohumeral ligament (MGL) disruption; craniomedial capsular lesions; and labral tears. Because MSS is usually a unilateral problem, the contralateral shoulder usually does not show signs of dysfunction, pain, or spasm on abduction.

Advanced Diagnostics. In addition to history, signalment, gait analysis, physical examination, orthopedic and neurologic examinations, and abduction angle tests, further diagnostic tests used to differentiate causes of shoulder pathology currently consist of hematology, biochemical profile, urinalysis, arthrocentesis, imaging modalities, and arthroscopy. Considered the gold standard, arthroscopic evaluation of the shoulder provides a definitive diagnosis of MSS or MSI.⁴⁷⁻⁴⁹ Arthroscopy of the shoulder allows for direct observation of all major intraarticular structures with magnification, "dynamic" evaluation of tissues during shoulder ROM tests, and "palpation" of intraarticular tissues using arthroscopic instrumentation.

Treatment

Patients are placed into one of three treatment categories; mild, moderate, or severe based on the results of the orthopedic examination, abduction angle tests, and arthroscopic scoring. Use of a shoulder support system/hobbles (such

as DogLeggs®)¹ and rehabilitation therapy is required in all cases.

Physical therapy for orthopedic conditions has a strong scientific background in human medicine. Many studies have shown the benefits of rehabilitation following rotator cuff surgery. The canine rehabilitation therapist should be aware of treatment techniques used in people and apply similar goals, principles, and techniques to dogs. Be aware that not all exercises work for every patient; some exercises require trial and error to achieve desired results. Some dogs tolerate the shoulder support system quite well, whereas others face a difficult time with the system. Some owners become frustrated with the shoulder support system and the restraint of the dog. Home education and owner compliance play a significant role in the postoperative management of MSS and MSI stabilization; therefore it is essential to reinforce the importance of the stabilization and the compliance of the program. Behavioral activities for the dog are beneficial.

Mild Medial Shoulder Syndrome (MSI). Patients with mild MSS most commonly have arthroscopic findings consisting of mild pathology (inflammation without fraying, disruption, or laxity of the MGL, subscapularis tendon, joint capsule) are placed in a shoulder support system/hobbles, such as DogLeggs®, and entered into a rehabilitation program. The rehabilitation program is restricted and consists of stabilization exercises only during the initial period. Cessation of activities, running, and jumping are required and leash walks must be enforced. The length of leash walks depends on the dog's endurance, pain, and inflammatory level. Quality over quantity should be emphasized with leash walks, and the owner should not push the dog to lameness with daily walks. Additional rehabilitative exercises include balance board, lifting of the opposite forelimb and hindlimb, weight shifts, standing on a balance ball, walking in figure eights and circles, and scapular stabilization work on a balance ball or disk. Stretching the shoulder in extension is permissible, but any abduction movements should be avoided. Modalities such as laser, cryotherapy, and moist heat are beneficial to reduce the inflammation and muscle spasms, respectively. Neuromuscular electrical stimulation may be used if there is a significant amount of atrophy and are usually targeted at the muscle belly of the supraspinatus and infraspinatus. The teres major is often restricted and painful as a result of the compensatory stabilization and may benefit from laser therapy and massage. Compensations in the cervical and thoracic regions should be addressed if present. In addition, underwater treadmill activity may be initiated. Typically water heights should be above the elbow to decrease the stress placed on the elbow. Swimming is only advised if the dog is able to swim in a controlled fashion without shoulder abduction. As the dog progresses and the weakness in the shoulder complex has improved, a gradual return to activity may

¹Shoulder support system, DogLeggs, Reston VA.



Figure 33-16 A, Dog wearing the shoulder support system following B, arthroscopic treatment for MSI.

be initiated. An increase in forelimb circumference should occur over time, along with increased active and passive shoulder extension, and decreasing lameness. This may take from 2 to 8 weeks depending on the dog, severity of injury, activity level, and commitment to rehabilitation. At this point a gradual return to activity may be initiated. The agility dog's jumping style and contact style may need to be adjusted.

Moderate Medial Shoulder Syndrome (MSS). Dogs with moderate MSS most commonly have arthroscopic findings that include subscapularis tendinopathy (disruption, fraying, core lesion), medial glenohumeral ligament pathology (fraying or disruption), craniomedial joint capsule focal synovial proliferation and/or disruption, and labral pathology (fraying, disruption, or tears). Additional findings occasionally include a bulge of the supraspinatus tendon with impingement on the biceps tendon. Depending on the tissues involved and the severity of disease, dogs in the moderate category are treated arthroscopically with a variety of techniques including radiofrequency-induced thermal capsulorrhaphy (RITC), suture imbrications (fiber wire), or stabilization using a specialized fiber band and buttons technique (TightRope®, Arthrex Co., Naples, FL).^{47,49,50} The current technique for radiofrequency treatment has been altered significantly since the previous report for the treatment of MSI in dogs⁵⁰ because of the detrimental effects that can occur with aggressive application. Currently a single pass striping technique versus multipass paintbrush technique has been developed based on current application in humans.⁵¹ In addition, a monopolar (vs bipolar) radiofrequency generator and probe using a default setting of 25 W and 70° C with high fluid flow egress has been employed to protect the surrounding tissues and cartilage. Radiofrequency-induced thermal capsulorrhaphy delivers radiofrequency energy to the capsule tissue, which causes the collagen to denature and the capsule to shrink. The optimal temperature to achieve

the most shrinkage without causing necrosis of the tissue is between 65° and 75° C. This treatment causes a significant decrease in mechanical stiffness for the first 2 weeks, and then, after the tissue undergoes active cellular repair from the surrounding uninjured tissue, the mechanical properties return to near normal by 6 to 12 weeks.⁵² To protect the tissues during this period dogs are placed in a shoulder support system/hobblers (such as DogLeggs®), such as those shown in Figure 33-16, for 3 months, and entered into a rehabilitation therapy program.

During the initial phases of the postoperative period, treatment is minimal and focuses on stabilization activities and PROM. Modalities aimed at reducing inflammation should not be used to maintain the inflammatory process. Active range of motion is not performed on the shoulder complex until 8 weeks postoperatively. Passive range of motion is performed on the digits, carpus, and elbow. Joint compressions may be performed to the digits, carpus, and elbow during the initial period, but the shoulder region should be avoided. If restrictions are present in the digits, carpus, or elbow, appropriate joint mobilizations to increase ROM may be performed.⁴ However, joint mobilizations should not be performed to the shoulder. Restrictions in the contralateral limb, spine, and hindlimbs should be evaluated and treated appropriately with ROM, joint mobilization, massage, laser, and other modalities. Leash walks are permitted but are restricted.

The hobblers must be worn at all times. Weight-shifting exercises may be added 4 weeks after surgery, as well as stretches of the cervical region. Sit-to-stands are also beneficial at this time. The dog should remain in the shoulder support system or hobblers at all times, with the exception of rehabilitation. Typically the dog is seen every other week for the first 8 weeks to monitor progress and ROM. At that point, weekly therapy is necessary and should consist of stabilization work, ROM, balance and proprioception exercises, and endurance

work. The underwater treadmill and more dynamic activities are not introduced until approximately 12 to 14 weeks postoperatively, although standing in water before that period is beneficial if the dog remains in its hobbles and does not thrash around in the water.

At 12 to 16 weeks postoperatively full ROM of the shoulder joint should be present without any soft tissue tenderness. Underwater treadmill therapy may begin at this stage. Also, appropriate modalities may be used if needed on any region. During the final stages of rehabilitation, approximately 12 to 16 weeks, the hobbles may be removed for controlled activity. Initially, short periods of exercise (5-15 minutes) are recommended without the hobbles and the activity should be controlled. For example, the dog may be off leash in a small fenced area. The goal is to remove the hobbles completely, unless the dog is in danger of slipping on the floor or during aggressive play. This may take several weeks to achieve. During rehabilitation, activities such as swimming, side stepping, cavaletti work, and hill walking may be added. All of the activities must be monitored and controlled. The activity may progress to tight turns, crawling activities, walking through tunnels, advanced balance activities, and increased endurance activities, such as leash walking and underwater treadmill.

The goals of the individual dog should be evaluated, and if the dog is returning to a sport such as agility, specific agility activities should be added 15 or 16 weeks postoperatively. Trotting, running, and jogging should be introduced on different surfaces and different intervals. Transitioning from a walk to a trot to a jog and back is an exercise that helps with proprioception and strength. Jumping can be introduced with a low-level jump in a straight line. The dog's normal jump height should be reduced by 50% to 75% depending on the height of the dog. For example, if the dog normally jumps 26 inches, the initial jump height should be 12 inches. Begin with one jump and progress to a series of three to five jumps in a row, spaced about three times the body length of the dog from each other. The dog should be jumped three to five times for three repetitions, two to three times per week. If no problems persist, the repetitions and the height can be gradually increased. Turns in the jumps may be added—setting consecutive jumps in a semicircle or a pinwheel. As the dog progresses, tighter turns can be added with a gradual return to normal activities.

Severe Medial Shoulder Syndrome (MSS) and Medial Shoulder Instability (MSI). Dogs with severe MSS or MSI typically have signs of craniomedial compartment laxity with shoulder abduction angles greater than 65 degrees. Arthroscopic findings usually include complete tears of the medial glenohumeral ligament and severe disruption of the subscapularis tendon, joint capsule and labrum. For this type of injury reconstruction of the medial compartment by direct tissue reapposition using synthetic capsulorrhaphy, bone/soft tissue anchors, and/or

TightRope® stabilization (Arthrex) may be indicated. Following surgical repair dogs are placed in a non-weight-bearing Velpeau sling for 2 weeks followed by a shoulder support system (such as DogLeggs®) for 2 to 3 months. Rehabilitation therapy is required for a longer period of time because recovery following primary reconstruction ranges from 4 to 6 months. Fortunately, severe cases are less common and are usually caused by trauma rather than repetitive activities, as seen in the mild and moderate categories.

Traumatic Fragmented Medial Coronoid Process

Traumatic fragmented medial coronoid process (TFMCP), sometimes called *jump down syndrome*, is a condition in the elbow joint of dogs that appears to occur commonly in performance dogs. Unlike the classic condition of fragmented medial coronoid process (FMCP), which affects the elbow joints of skeletally immature large to giant breed dogs, TFMCP appears to have no age or size limitations.

Anatomy

The elbow joint as a whole is a trochlea arthrosis or ginglymus (hinge) joint, allowing flexion and extension of the humerus over the radius and ulna. In dogs, the caudal aspect of the head of the radius rotates in the radial notch of the ulna during pronation and supination.²² The normal elbow joint is characterized by a smooth transition from the ulnar articular surface to the radial surface. The medial coronoid process of the ulna is level with or slightly below the surface of the radius (Figure 33-17).

Cause

The cause of TFMCP is poorly understood. It is possible that abnormal repetitive loading, such as landing from a jump, hitting contacts or a flyball box, may lead to microfractures of the bone underneath the cartilage (subchondral fractures).⁵³ Additionally, increased repetitive loading can arise from contraction of the biceps/brachialis muscle

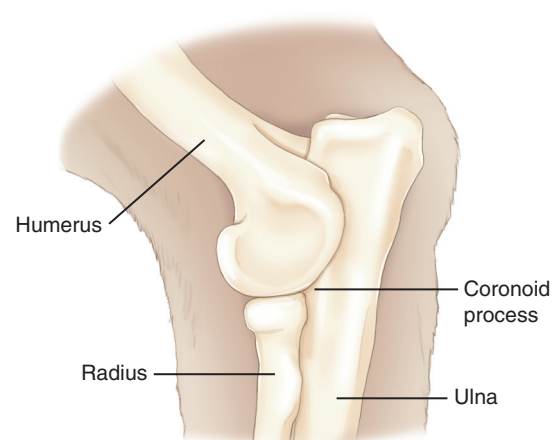


Figure 33-17 Normal canine elbow anatomy.

complex. When the biceps/brachialis contracts, a force is generated that rotates the medial coronoid into the radius. These microcracks disturb the mechanical properties of bone, and if not repaired properly through normal healing mechanisms, fatigue fractures develop. Additionally, loss of osteocytes (bone cells), indicated by decreased osteocyte density, has been strongly associated with the presence of microdamage after fatigue loading.⁵³ These studies imply that excess load may lead to fatigue microdamage of the subchondral trabecular bone and eventual fracture, which may play an important role in the pathogenesis of TFMCP. Dogs may be further predisposed to this condition if they have elbow dysplasia. Dogs with elbow dysplasia may have asymmetric growth of the radius and ulna during development, resulting in elbow joint incongruity. Elbow incongruity, such as radioulnar step defects, humeroulnar incongruence/conflict, and varus deformity of the humerus, causes abnormal contact patterns in the elbow, specifically at the coronoid-trochlear articulation, which is theorized to increase the load on the medial coronoid process (MCP).⁵⁴ Regardless of the etiology, if left untreated as a continual lameness, secondary osteoarthritis may progress, as noted by damage to the cartilage such as softening, fibrillation, fissuring, and erosions as well as additional subchondral bone microcracks and fragmentation. The free fragments may contribute to frictional abrasion (“kissing lesions”) of the opposing surface, including the medial aspect of the humeral condyle and radial head.

Diagnosis

History and Clinical Signs. Dogs with TFMCP may be presented with a history ranging from a subtle intermittent off-loading of the forelimb to significant unilateral or bilateral forelimb lameness. This lameness is typically exacerbated with exercise and heavy activity. The onset of lameness is insidious. As lameness persists, it may increase in severity. Affected dogs often place the carpus in an exaggerated valgus position (turned out) when sitting or standing, and circle the foreleg outward and move the elbow away from the midline (circumduct the antebrachium and abduct the elbow) during the swing phase of the stride. The history of dogs with TFMCP typically includes a lack of response to rest and NSAIDs. Many dogs with TFMCP are mistakenly treated for shoulder pathology because the attending veterinarian elicits a pain response when the shoulder is extended. The authors believe the pain response actually arises from the elbow because when the veterinarian performs an extension maneuver of the shoulder, the elbow is usually simultaneously extended. Extension of the elbow causes tension in the biceps/brachialis muscle complex. Tension in the biceps/brachialis exerts pressure on the medial coronoid and overlying inflamed joint capsule causing the pain response.

Orthopedic Examination. On physical examination discomfort is usually noted on direct palpation of the medial compartment of the elbow joint, specifically the medial coronoid process. Discomfort may also be noted on hyperflexion of the elbow. Most dogs with TFMCP are reluctant to allow full elbow flexion. In chronic cases, full flexion may not be obtained. Crepitus may be noted when placing the elbow through ROM. Some investigators suggest that the carpus should be placed in a flexed, externally rotated position while the elbow is extended to exacerbate the pain response. Joint effusion may be detected as a fluctuant swelling distal to the lateral or medial epicondyle of the humerus. Depending on the chronicity, atrophy may be noted in the affected forelimb muscles.

Advanced Diagnostics. In addition to history, gait analysis, physical examination, and orthopedic and neurologic examinations, further diagnostic tests used to differentiate causes of elbow pathology currently consist of hematology, biochemical profile, urinalysis, arthrocentesis, imaging modalities, and arthroscopy. Unfortunately, radiographs may be of little value in early cases because of difficulty identifying the fragment or line of separation using standard radiography.⁵⁵ In chronic cases, however, radiographs may reveal secondary evidence of bony remodeling consistent with osteoarthritis. These changes may include sclerosis of the ulnar notch and remodeling along the anconeal process and MCP.⁵⁶ Advanced diagnostic imaging modalities such as CT scans, MRI, nuclear scans, and arthroscopy may allow confirmation of the condition.⁵⁶ Arthroscopic evaluation of the elbow joint allows direct observation of all major intraarticular structures with magnification, “dynamic” evaluation of tissues during elbow ROM tests, and “palpation” of intraarticular tissues using arthroscopic instrumentation.⁵⁷ Arthroscopic exploration of the elbow provides a definitive diagnosis of TFMCP when a fragment or cartilage fissure is observed (Figure 33-18). In a small percentage of cases advanced imaging (nuclear scan, CT, MRI) may indicate fragmentation of the MCP, but arthroscopic observation does not reveal a fissure or fragment. In such cases the fissures (microcracks) are believed to be within the coronoid process beneath the cartilage surface.

Treatment

Treatment of TFMCP is multimodal and includes a combination of medical and surgical management as well as rehabilitation therapy. Through a multimodal approach it may be possible to relieve pain and maintain limb function, as well as to return the dog to a near-normal level of activity and competition.

Arthroscopy is an excellent modality for diagnosing TFMCP as well as a minimally invasive means of treatment. Arthroscopic removal of the fragments is recommended not only to remove the inciting cause of lameness



Figure 33-18 Arthroscopic image of a traumatic fragmented medial coronoid process in an agility dog.

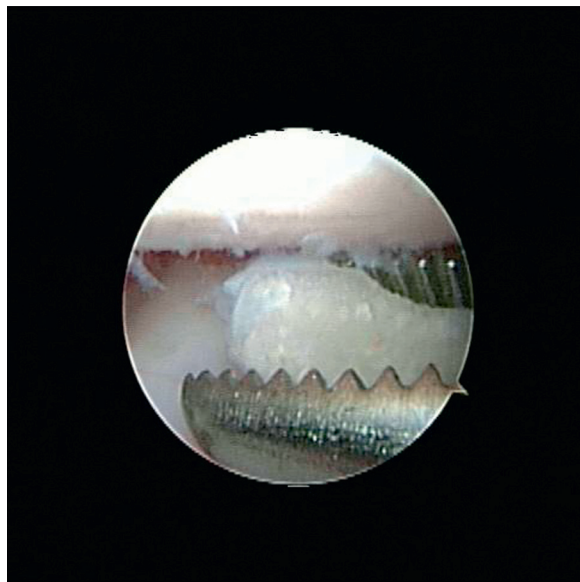


Figure 33-19 Arthroscopic treatment of a traumatic fragmented medial coronoid process.

but also to help slow the progression of osteoarthritis. Arthroscopic treatment may include a combination of techniques such as fragment removal, débridement of diseased tissues, creation of vascular access by abrasion arthroplasty, forage, microfracture, and subtotal coronoid osteotomy depending on the progression and severity of disease (Figure 33-19).⁵⁸ Arthroscopy allows for superior visualization of structures in the joint and is also believed to have less soft tissue trauma, shorter surgery and hospitalization

times, decreased risk of infection, and shorter recovery times compared with the traditional surgical approach of fully opening the joint (arthrotomy).

Following arthroscopic treatment, dogs are typically prescribed NSAIDs for 14 days to help decrease inflammation and discomfort. Polysulfated glycosaminoglycan (PSGAG), an injectable joint protective agent, is also recommended following arthroscopic treatment and is administered intramuscularly twice a week for up to 4 weeks. Polysulfated glycosaminoglycans have been shown to decrease the amount of degradative enzymes (which stimulate inflammation and cartilage erosion) as well as promote repair and regeneration of cartilage.⁵⁹ In addition, an oral joint protective agent such as glucosamine, chondroitin sulfate, and avocado/soybean unsaponifiables (ASU) is recommended as a daily supplement for life. The antiinflammatory properties and pain-reducing effects of these supplements have been studied.^{60,61} There is also evidence that they may have a chondroprotective effect.^{60,61}

As with any rehabilitation therapy program, rehabilitation after elbow arthroscopy should follow a sequential and progressive multiphased approach.⁶² The ultimate goal of elbow rehabilitation is to return the patient to its previous functional level as quickly and safely as possible. However, each case is uniquely different, and several factors—including duration of injury/lameness before surgery, secondary osteoarthritis and surgical intervention—all influence the rate at which the patient proceeds through the healing process.

The elbow is predisposed to flexion contractures because of the intimate congruency of the joint articulations, the tightness of the joint capsule, and the tendency of the cranial capsule to develop adhesions following injury. The biceps/brachialis complex also attaches to the capsule and crosses the elbow joint before becoming a tendinous structure and inserting on the ulna. Injury to the elbow may cause excessive scar tissue formation of the brachialis muscle as well as functional splinting of the elbow. Reestablishing full elbow extension is the primary goal of early ROM activities to minimize the occurrence of elbow flexion contractures.⁶³⁻⁶⁵

Additional goals of this stage are to protect healing tissue, decrease pain and inflammation, restore weight bearing, and retard muscular atrophy.⁶⁶⁻⁷⁴ The rehabilitation therapist must not overstress healing tissues during this phase. It is important that the owner restrict the dog's activity to short leash walking only with no running, jumping, and rough play.

Cryotherapy (Figure 33-20) may be performed during the initial inflammatory period to assist in reducing pain and inflammation. The primary physiologic effects include vasoconstriction, reduced cellular metabolism and permeability, decreased motor and sensory nerve conduction velocity, analgesia, and decreased muscle spasm.⁷⁵⁻⁷⁷ Ice

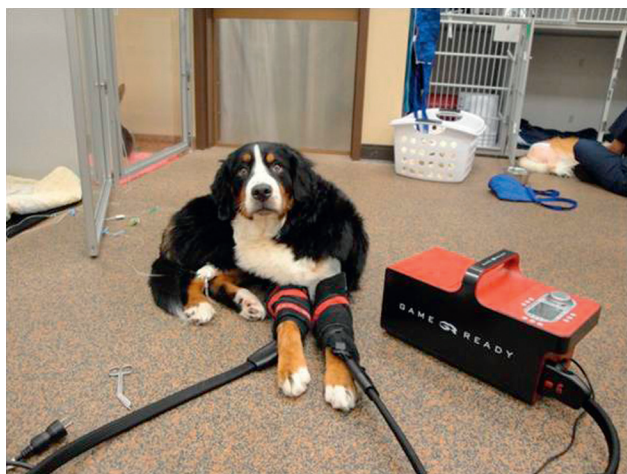


Figure 33-20 Cryotherapy of the elbow immediately post arthroscopy.



Figure 33-21 Range of motion of canine elbow following arthroscopic treatment.

may be applied around the entire elbow joint for a duration of 10 to 20 minutes, three to six times daily.

Range of motion is performed to the entire forelimb: toes, carpus, elbow, and shoulder (Figure 33-21). Slow controlled movements are performed with slight over pressure at end range. Range of motion activities are usually performed two to three times daily. Range of motion activities are performed for all planes of elbow and carpus motions to prevent the formation of scar tissue and adhesions. Grade I and II joint mobilizations (Figure 33-22) may be performed during the early phase of rehabilitation as tolerated.⁷⁸ Joint mobilizations increase circulation, proprioception, and healing and may decrease swelling. Pain-free grade III posterior glides with oscillations are performed in the midrange of motion to assist in regaining full elbow extension. Aggressive mobilization techniques are not used until later stages of rehabilitation when pain has subsided. Grade I and II mobilization techniques are also used to neuromodulate pain by stimulating articular receptors.^{78,79} Grade III and IV joint mobilizations also should be performed to all joints that are hypomobile or restricted in any direction. Common regions of decreased mobility may also include the scapula, cervical/thoracic spine, or pelvis.

Manual therapy, including soft tissue massage/mobilization and stretching of the surgical forelimb, non-surgical forelimb, cervical spine, and hindlimbs may be necessary because of compensations that occur before and/or secondary to surgery. If the dog continues to have difficulty achieving full extension using ROM and mobilization techniques, a low load, long duration (LLLD) stretch may be performed to produce a creep of the collagenous tissue, which results in tissue elongation.⁸⁰⁻⁸³ The dog lies in lateral recumbency with a towel roll placed under the brachium to act as a cushion and fulcrum. Light pressure

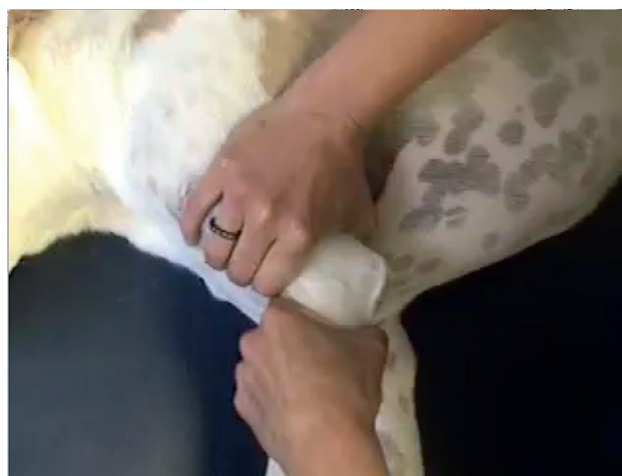


Figure 33-22 Joint mobilization of the elbow in a dog following arthroscopic treatment.

is applied to the carpus and distal antebrachium. The amount of pressure applied should be of low magnitude to enable the dog to perform the stretch for the entire duration without pain or muscle spasm. The aggressiveness of stretching and mobilization techniques is dictated based on healing constraints of involved tissues as well as the amount of motion and end feel. If the dog has decreased motion and a hard end feel without pain, aggressive stretching and mobilization techniques may be used. Conversely, a dog exhibiting pain before resistance or empty end feel should be progressed slowly with gentle stretching.

Several modalities may be used to aid in the healing process during this stage and serve as an adjunct to manual treatments and therapeutic exercises. Laser therapy (Figure 33-23) mediates a variety of events including release of



Figure 33-23 Laser therapy of the elbow following arthroscopic treatment.

growth factors, cytokine reactions, and cell replication; the ultimate effect of these events is accelerated tissue healing.^{84,85} Laser treatment dosage is provided in Joules (energy). A typical treatment of 4 to 6 Joules per cm² is applied covering the entire treatment area. Laser therapy can be performed daily for the first week, and then decreased to every other day during week two. After week two, laser may be applied on an as needed basis.

Ultrasound therapy is a commonly used modality in human medicine.⁸⁶ Ultrasound has a direct effect on cells and in laboratory studies, to stimulate healing. A treatment schedule may include daily treatment for up to 10 days, followed by less frequent sessions as the condition improves. Continuous wave ultrasound produces heating of the tissues, which may be beneficial before stretching.

Transcutaneous electrical nerve stimulation (TENS) (Figure 33-24) is used commonly for pain relief. The primary effect of pain relief is obtained through two main mechanisms: the inhibition of the pain via the gate theory, and via descending inhibitory mechanisms (release of endogenous opiates).⁸⁷ A recent study in dogs showed significant improvement in ground reaction forces (weight bearing) immediately after treatment that lasted for up to 3 hours.⁸⁸ Treatments initially may be applied three to seven times per week and then decrease in frequency as pain and lameness subside.

After the acute inflammatory phase has passed, moist heat, warm whirlpool, and ultrasound may be used at the onset of treatment to prepare the tissue for stretching and to improve the extensibility of the joint capsule and musculotendinous structures.

The early phases of rehabilitation must also focus on retarding muscular atrophy through therapeutic exercises. Alternating rhythmic stabilization drills for forelimb flexion/extension/horizontal abduction/adduction and shoulder



Figure 33-24 Transcutaneous electrical nerve stimulation of the elbow following arthroscopy.



Figure 33-25 Three-legged standing.

internal/external rotation are performed to begin reestablishing proprioception and neuromuscular control of the upper extremity.^{89,90} Leash walking may be initiated after surgery, beginning with 5 to 10 minutes, three times a day, with increasing duration to 20 minutes as tolerated, by the end of the acute phase. Other weight-bearing exercises are initiated at this time and are performed on a daily basis. Some examples include the following:

- Leg standing—Lift nonsurgical forelimb and shift weight to surgical limb (Figure 33-25).

- Down to stand—Ask patient to lie down and then to stand. Use treats as needed.
 - Rocking on all fours—Dog stands on all four limbs. Gently rock the dog from side to side and diagonally, with 3-cm perturbations in each direction. Promotes weight bearing and proprioception to the affected limb.
- When the dog exhibits full ROM, minimal pain and tenderness, and improved lameness progression, the goals should be focused to enhance elbow and upper extremity mobility, improve muscle strength and endurance, and reestablish neuromuscular control of the elbow complex.

Stretching exercises are continued to maintain full elbow flexion and extension. Mobilization techniques may be progressed to more aggressive grade III and IV techniques, as needed, to apply a stretch to the capsular tissue and improve end range. Flexibility progresses during this phase to focus on the carpus, elbow, and shoulder, including—flexion, extension, internal and external rotation.

Also during this stage emphasis is placed on increasing strength and proprioception of the affected limbs. Strength exercises may include longer leash walks, including up and down hills, stepping over objects such as poles/sticks, and walking over uneven surfaces such as sand, mulch, or tall grass. Initiate walking in circles and figure eights by starting with moderately sized circles, and decrease the diameter weekly until tight circles are achieved. Walking in a figure eight pattern may be done around cones placed 5 to 6 feet apart depending on the size of the dog. Additional therapeutic exercises include the following:

- Ladder—Use an extension ladder on a level floor or cavaletti rails. Walk the dog slowly through ladder or rails to ensure individual weight bearing on all limbs.
- Wheelbarrow exercises—Elevate both rear legs and have the dog walk forward on the front limbs only. Begin for short distances and increase as tolerated.
- Hydrotherapy—The use of underwater treadmill or swimming may be effective during this stage as well. NOTE: Full ROM of the elbow should be present and there should be no palpable tenderness/inflammation of the elbow or shoulder. Hydrotherapy sessions are performed two to three times/week and are initiated slowly, beginning with 5 minutes, and increased 1 to 2 minutes per session as tolerated.

Neuromuscular control exercises are initiated in this phase to enhance the muscles' ability to control the elbow joint during athletic activities. These exercises include proprioceptive neuromuscular facilitation exercises with rhythmic stabilizations and slow reversal manual resistance elbow/carpus flexion exercises. One method includes having the dog weight bearing and manually shifting the dog back and forth from a cranial to caudal direction on varying surfaces, and ambulating on uneven surfaces while applying manual forces.

The next phase involves a progression of activities to prepare the dog for return to off-lead activity and sport participation. The goals of this phase are to gradually increase strength, power, endurance, and neuromuscular control, to achieve a gradual return to sport. Specific criteria that must be met before entering this phase include full nonpainful ROM, no pain or tenderness, and strength and muscle mass that is 70% of the contralateral forelimb.

Advanced strengthening activities during this phase include exercises emphasizing high-speed and eccentric muscle contraction and plyometric activities. Elbow flexion exercises using elastic bands (Thera-Bands™) are progressed to emphasize eccentric control. Elbow flexion can be performed with elastic bands to emphasize slow and fast speed concentric and eccentric contractions. The elastic bands are placed on the cranial aspect of the mid humerus and the practitioner stands behind the dog while ambulating to offer eccentric resistance.

Plyometric drills are an extremely beneficial form of exercise for training the upper extremity musculature.⁷⁴ The physiologic principles of plyometric exercise use an eccentric prestretch of the muscle tissue, thereby stimulating the muscle spindle to produce a more forceful concentric contraction. Plyometric exercises are performed using jumps during the later stages of this phase to train the upper extremity musculature to develop and withstand high levels of stress. Examples of these activities include the use of a single jump exercise, then a series of jump grids consisting of four jumps, initially placed in a straight line, then in an arc, and progressing to a serpentine or threaded configuration. Another example is simply jumping on and off a box at several different angles.

Increasing time, distance, speed, or terrain during leash walks aids strengthening. Steeper hills and stairs may aid in advanced strengthening. Proprioception and stabilization exercises may be advanced with the use of wobble boards, BOSU®, and/or therapeutic ball exercises (Figure 33-26). Placing the dog on an unstable surface such as these further challenges the patient's proprioception as well as improves strength of the proximal forelimbs.⁹¹ Progression of these exercises may include gentle rocking from side to side and having the patient follow a treat in all directions. Eccentric exercises such as walking through a tunnel and walking backward also may be added. This will challenge the cranial muscles of the shoulder and elbow.

The final phase of elbow rehabilitation, the return to activity phase, allows the dog to return progressively to full off-lead activity and competition using an interval training return to sport program. Sport-specific functional exercises are performed to prepare the dog for the stresses involved with each particular activity or sport.

Before a dog is allowed to begin the return to activity phases of rehabilitation, the dog must exhibit full ROM, no pain or tenderness, have equal forelimb muscle



Figure 33-26 Proprioception and stabilization exercises may be advanced with the use of a BOSU.

symmetry, no signs of lameness at a walk, trot, or when performing tight circles at a trot, and a satisfactory clinical examination.

If healing of all tissues has occurred appropriately, advanced strengthening techniques may be encouraged such as jumping (on/off a couch or over a jump). Longer walking or trotting sessions may be necessary to build up strength and cardiovascular endurance. Gradual off-leash activities may be initiated at this time. The duration of exercises may be increased, and additional manual balance perturbations in all directions may be initiated on uneven surfaces, such as wobble boards (Figure 33-27), BOSU, or therapy ball. Lifting one limb while performing these activities further increases the difficulty.

After achieving the previously mentioned criteria to return to sport, a formal interval sport program is initiated.⁹² For the performance dog, training is initiated three times per week with a day off between sessions. Each step increase in activity is performed at least twice on separate days before the dog is allowed to progress to the next step. Full activity should be performed without pain or significant increase in clinical signs (lameness, joint effusion, and discomfort on ROM). If the dog experiences worsening clinical signs at a particular step of the program, the owner/trainer is instructed to regress to the prior step until the signs subside. It is important for the performance dog to perform active stretching before training and performance and to do mild warm-up activity, such as jogging. This provides an adequate warm-up, but also ensures maintenance of necessary ROM and flexibility of the forelimb.

Advanced strengthening and proprioception must be accomplished if return to sporting activities is a goal. Advanced training and rehabilitation may be accomplished with activities such as plyometrics, sprinting, jumping,

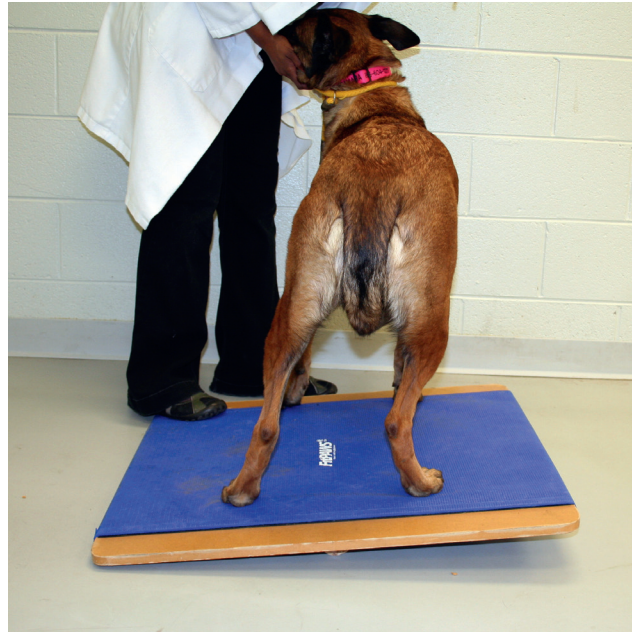


Figure 33-27 Manual balance perturbations in all directions are initiated on uneven surfaces such as a wobble board.

agility training, and sport-specific training. Greater balance perturbations on the BOSU or therapy ball; quick directional changes, pivoting, figure eight exercises at submaximal effort initially, then progressing to greater speeds; cardiovascular endurance; and a continuation of strength training should be targeted.⁹³

Hindlimb Conditions

The most common hindlimb orthopedic/sports medicine conditions afflicting performance and working dogs are iliopsoas muscle strain and cranial cruciate ligament (CCL) insufficiency. Cranial cruciate ligament insufficiency is presented elsewhere in this text. Presented in the following are the most common muscular injuries/conditions of the hindlimb: iliopsoas strain and gracilis and semitendinosus/semimembranosus contracture. Iliopsoas strain is seen much more frequently than gracilis or semitendinosus/semimembranosus contracture. Because contracture of the gracilis or semitendinosus/semimembranosus is less common, it is often undiagnosed until significant pathology is established. Diagnosis and treatment in the early stages of the disease process can significantly inhibit the progression, emphasizing the need for early identification.

Iliopsoas Strains

Acute, Stretch-Induced Injuries

Iliopsoas strain is an overstretching injury. It is estimated that muscle injuries account for more than 30% of all

injuries seen in a typical human sports medicine practice, and has been reported to be the most common injury seen in human general practices.^{7,94} Laboratory reports indicate that muscles that cross multiple joints or that have complex architecture have increased susceptibility to strain injury.⁷ Strain injuries result from either excessive stretching or stretching when the muscle is being activated.⁹⁴ The injured muscle is weaker and has reduced tension, increasing the risk of reinjury.^{7,94} Appropriate treatment and allowing for complete healing is a imperative to prevent reinjury. Rehabilitation therapy initially focuses on relieving pain and inducing healing followed by muscle strengthening, endurance and flexibility. A complete program should also include educating owners on proper warm-up and stretching techniques to further reduce the incidence of reinjury.

Anatomy

The psoas major and the iliacus muscles combine to form the iliopsoas muscle (Figure 33-28). The psoas major muscle arises from the transverse processes of the lumbar vertebrae at L2 and L3 and the bodies of L4-7, and the iliacus muscle arises from the ventral surface of the ilium. They have a common insertion on the lesser trochanter of the femur. The action of this muscle is to move the pelvic hindlimb forward relative to the trunk via hip flexion.²²

Cause

Iliopsoas muscle strains occur as the result of excessive force acting on the muscle, and are commonly associated

with highly athletic activities such as agility.⁹⁴ These injuries often occur at or near the muscle-tendon junction. Eccentric contraction, in which the muscle is contracting while it is stretched, is thought to be an important factor in the development of these acute strain injuries.⁹⁵ Traumatic incidents that result in active eccentric muscle contraction, such as slipping into a splay-legged position, jumping out of a vehicle, aggressive agility training, or roughhousing with other dogs, are often reported to precipitate acute lameness. It is not uncommon to find concurrent orthopedic problems or a recent history of surgical treatment for another orthopedic condition, such as cranial cruciate ligament rupture.

Diagnosis

The history for dogs with iliopsoas strains commonly includes a lameness ranging from a subtle intermittent off-loading of the hindlimb to significant unilateral hindlimb lameness that is exacerbated with activity. These dogs commonly demonstrate performance issues such as knocking down the bars of jumps with the hindlimbs or slowing in the weave poles.

On direct palpation, tightness, discomfort, and spasm of the myotendinous unit may be noted (Figure 33-29). Pain and spasm may also be noted when stretching the myotendinous unit by either placing the hip in extension with abduction or by simultaneous extension of the hip with internal rotation of that pelvic hindlimb (Figure 33-30).

Radiographs are of little value in the early phase of the condition but may reveal mineralization just cranial

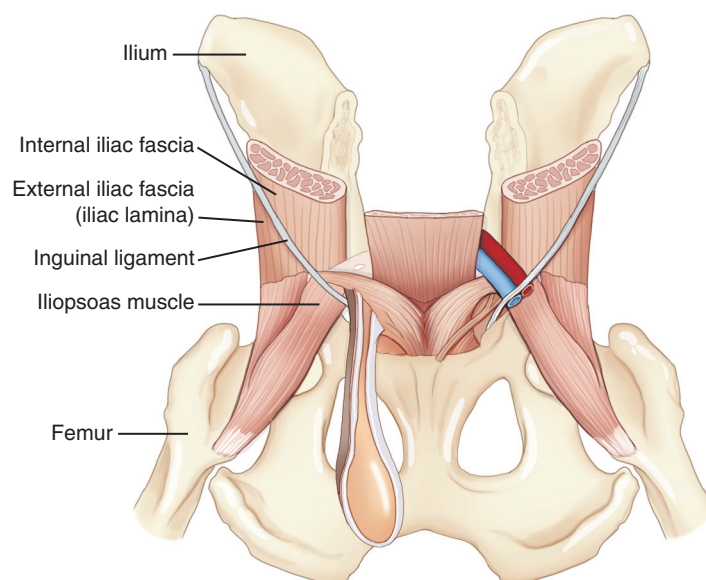


Figure 33-28 Iliopsoas anatomy.



Figure 33-29 Palpation of the iliopsoas.



Figure 33-31 Laser therapy for an iliopsoas strain.



Figure 33-30 Iliopsoas passive stretch.

to the lesser trochanter in chronic cases. The use of advanced imaging modalities to demonstrate lesions of the affected muscle and/or tendon can increase confidence in the diagnosis. Ultrasonography is a relatively inexpensive noninvasive imaging modality for canine musculoskeletal evaluation with the additional advantages that general anesthesia is not required and response to treatment can be monitored with follow-up imaging. This imaging modality depends on the expertise of the operator, which may limit its practical application in some settings. Advanced diagnostics such as CT scan and MRI may be used to identify iliopsoas strains and both are widely used to diagnose acute, stretch-induced muscle injury in human patients.⁹⁶ Although CT is valuable for imaging soft tissue lesions, the use of MRI has greatly increased the ability to detect subtler lesions.

Treatment

Acute iliopsoas strains should be treated conservatively. Skeletal muscle relaxants may be administered in severe cases to reduce pain and muscle spasms. Medical management may also include NSAIDs, cryotherapy, and controlled activity. Rehabilitation can be very effective in treating iliopsoas strains. Treatments may include laser therapy to increase circulation, remove waste products, and promote healing (Figure 33-31). Pain-free PROM and high-repetition exercise also are recommended. Acute strain injuries should not be stretched because microtears may occur. Core strengthening exercise is essential in the return of the athletic dog to competition and a pain-free lifestyle. Exercises using a therapy ball should be performed to increase the dog's core muscles, including the lower back and abdominal muscles. Because the iliopsoas originates from the lumbar spine, the lumbar spine and lumbosacral area may need to be treated in acute cases. Passive range of motion, interferential electrical stimulation, stabilization exercises, and gradual increases in weight bearing of the involved limb are recommended. Active ROM and strengthening exercises are added next. Strengthening exercises might include stepping over cavaletti poles, hemi-standing (lifting both front and hindlimbs on one side of the dog's body while balancing on the contralateral limbs), standing with the front paws on an elevated surface, and use of the wobble board. Hill walking should be added to increase the strength of the pelvic limb musculature. A steep hill of approximately 20 to 40 degrees in incline should be used to slowly walk the dog up the hill. It is important that the dog walks, and does not bunny hop or trot, up the hill. Slow walking promotes an equal balance of each hindlimb and focuses on the hamstring and the gluteal muscles. The dog should be walked up the hill approximately 100 feet and then slowly walked down.

The downward motion works the iliopsoas muscle in an eccentric fashion. This can be repeated three to five times and repeated a few times during the week. Acupuncture may be helpful to assist with pain control and to promote healing, including the lumbar and lumbosacral region. Joint mobilization and other manual therapy may also be needed to assist with lumbar spine ROM and motion of the coxofemoral joint.

In chronic iliopsoas strains it is important to reinstate the inflammatory process to assist in remodeling of the tendon fibers. Nonsteroidal antiinflammatory drugs should be avoided with chronic iliopsoas strains as they impair the inflammatory response. Rehabilitation therapy is recommended with chronic iliopsoas strains. Modalities might include heat, ultrasound, and laser, followed by massage therapy. Be sure to check for pain or discomfort in the lumbosacral joint and lumbar region (lower spine) during manipulation. Chronic iliopsoas strains may arise from a problem with mechanics; therefore working on correcting the mechanics of movement may help to reduce the strain on the iliopsoas and contribute to its healing. If movement mechanics are not corrected, the condition may not improve. The exercise progression is similar to that for acute iliopsoas strains, but initiating stretching (hip extension with abduction) after modalities and massage is advised, as are longer walks. In chronic muscle strain injuries it usually takes longer to recover and progress through the stages of healing and exercise because of the chronic nature of the changes in the myotendinous unit. Education of owners/trainers is extremely important because they should be instructed to move the dog slowly through the phases of recovery. Core strengthening and activities focusing on strengthening of the gluteals and hamstrings, as well as the eccentric strength of the iliopsoas, should be emphasized. The underwater treadmill, hiking, and hill work are all appropriate. With regard to aquatic therapy, swimming may aggravate the iliopsoas injury because it forces the body to maintain the hip in a shortened or flexed position with more rapid movement of the limbs than occurs with walking on an underwater treadmill.

When returning to agility training, weave poles and tight turns at full jump heights should be avoided during the early stages of retraining. The risk of reinjury to a previously strained muscle is well established in human patients when a previous minor injury often predates a major strain injury. The risk of more significant strain is increased when preexisting strain injury has not completely healed. This also appears to be the case in canine patients, in whom pain from iliopsoas strain injuries may recur. Appropriate warm-ups, stretching, and retraining are extremely important in preventing injury and in returning the dog to a competitive performance level. Examples of retraining techniques for iliopsoas strains include starting with low, straight-line jumps, and later incorporating very wide sweeping turns that progress over time to higher

jumps with tighter angles. Do not include higher jumps and weave poles until late in the retraining period. It has been deduced from human rehabilitation that stretching without an appropriate warm-up is detrimental to both the dog's tissues and its performance. In fact it is much more important to do an appropriate warm-up than stretching if time does not permit both. Following training and performance, make sure to implement iliopsoas stretching techniques (hip extension with abduction) and use an appropriate cool-down with ice therapy (5 minutes on, 5 minutes off, 5 minutes on).

Regenerative medicine therapy consisting of a combination of stem cells and platelet rich plasma may be considered for those cases refractory to conservative medical management and rehabilitation therapy. Injections should be performed with ultrasound guidance to ensure proper placement of cells. This author has had success with this treatment followed by an appropriate rehabilitative therapy program. In more severe cases, additional injections may be needed.

Surgical treatment may be warranted for those that do not respond to conservative medical management and rehabilitation therapy. In cases where there are irreversible changes to the myotendinous unit, such as fibrosis (forming excess fibrous tissue while healing) of the muscle-tendon junction, surgical treatment by tenotomy/tenectomy (releasing the tendon) or reattachment may be indicated.⁹⁷ Surgical intervention should also be considered when the strain injury recurs at regular intervals, although the lesion should be first confirmed with ultrasonography, CT, or MRI imaging. Good to excellent results have been reported with dogs returning to function, although performance dogs may work at a lower level.

Gracilis and Semitendinosus/Semimembranosus Myopathy/Contracture

Muscle contracture is defined as "The abnormal shortening of muscle tissue, rendering the muscle highly resistant to stretching."⁹⁸ A contracture can lead to permanent disability.⁹⁸ There is a significant difference between muscle contraction and muscle contracture. *Contraction* refers to the normal physiologic process of muscle shortening, resulting in work. *Contracture* refers to the abnormal pathologic process resulting in fibrosis and permanent damage to a muscle. It is important to remember that a muscle can have a significant amount of scar tissue from prior injury and still not be considered a contracture based on its ability to function. The pathologic process of muscle contracture is characterized by replacement of most or all of the entire muscle and/or associated tendon with fibrous connective tissue. This process can take weeks to months and leads to shortening of the affected muscle or muscle group. A muscle contracture can also severely affect the associated joints by fixing them in flexion or extension. In most cases, response to treatment and

prognosis is better for muscle contractures of the forelimb than the hindlimb.

Canine muscle contracture is reported to affect several different muscles, is associated with a number of predisposing factors, and has a varied prognosis depending upon which muscle is affected. Most patients suffer some form of trauma weeks to months before the contracture is present. The clinical signs include lameness, pain, weakness, decreased ROM, firmness noted throughout the entire muscle, and usually a characteristic gait. Predisposing factors for muscle contracture include compartment syndrome, infection, trauma, repetitive strains, fractures, infectious diseases, immune-mediated diseases, neoplasia, and ischemia. Contractures of the gracilis and semitendinosus/semimembranosus may occur individually or concurrently. History associated with a contracture of the gracilis and/or semitendinosus/semimembranosus typically consists of a single or repeated strain injury⁹⁹ with a gait abnormality developing after. Gait abnormalities usually start abruptly and progress over a period of 6 weeks to months,^{40,99} at which time the condition becomes static. Locomotion following gracilis contracture is characterized by raising the affected leg in a jerky fashion with the hock hyperflexed and rotated laterally, the metatarsus rotated medially, and the overall stride shortened.^{40,99} A shortened stride with a rapid medial rotation of the paw, external rotation of the hock, and internal rotation of the stifle mid-stride is noted with contracture of the semitendinosus.⁹⁹ There appears to be some breed and age predilection; however, the sex of the animal does not have an appreciable influence. Gracilis muscle and semitendinosus/semimembranosus contracture most often affects highly active German shepherd dogs and shepherd-related breeds between the ages of 3 and 7 years.^{40,99,100} The suspected cause in the working and performing German shepherd is repetitive strain injury leading to secondary contracture.

Anatomy

The gracilis is a broad muscular sheet that arises from the pelvic symphysis, extends superficially along the caudal aspect of the inner surface of the thigh and tapers to its end along the entire length of the cranial border of the tibia. An aponeurosis expands out into the crural fascia. From the caudal border, a well-developed reinforcing band extends to the calcanean tendon with attachment to the tuber calcaneus. The gracilis is responsible for adduction of the thigh, extension of the hip, flexion of the stifle, and extension of the hock.

Arising from the tuber ischium, the semitendinosus inserts distally on the medial surface of the tibia, with a second insertion by means of a fascial attachment to the tuber calcaneus. It is responsible for extension of the hip and tarsal joints and flexion of the stifle joint in the free non-weight-bearing limb. The semimembranosus also arises from the tuber ischium then extends distally with

insertion on the caudal surface of the femur distal to the pectineus, and to the proximo-medial aspect of the tibia. It is responsible for extension of the hip and flexion of the stifle joint in the free non-weight-bearing limb.

Cause

Skeletal muscle tissue cells have a very limited ability for regeneration.¹⁰¹ Repair to damaged muscle tissue begins early in the inflammatory cascade and involves two processes: regeneration of injured tissue by parenchymal cells of the same type, and replacement by connective tissue (fibroplasia).¹⁰¹ The balance of the two dictates the amount of new skeletal muscle tissue or scar tissue produced. It appears to be this limitation for skeletal muscle cell regeneration that drives the formation of scar tissue and inevitable fibrosis after skeletal muscle trauma instead of regenerating new cells to replace the damaged muscle.

The exact cause of fibrotic myopathy is usually unknown. The fibrotic myopathy may result from acute trauma, chronic repetitive trauma, autoimmune disease, drug reactions, infections, neurogenic disorders, and vascular abnormalities.¹⁰² Fibrosis and contracture may also result from ischemia secondary to indirect trauma.^{102,103} Histologic examination reveals dense, collagenous connective tissue in place of muscle.¹⁰²

Muscle strains can be characterized by tearing of muscle fibers, most often at or near the musculotendinous junction, that result from excessive force or stress to the muscle.¹⁰⁴ Indirect muscle injuries in humans often occur subsequent to rapid acceleration during athletic activities.¹⁰⁴ The type and severity of injury directly correlates to how healing will occur; predominantly by regeneration of functional myofibrils or scar tissue formation.⁹⁷ Fibrosis and contracture with minimal regeneration results when muscle damage is severe. Fibrous scar tissue provides tensile strength and is involved in the natural healing process of muscle tissue, however, when excessive, scar tissue limits regenerative healing and interferes with muscle contraction and relaxation resulting in mechanical lameness of varying degrees.^{99,100,103,105,106}

Susceptibility of this disorder appears greater in active dogs. Therefore it is reasonable to suspect the same mechanism of injury may play a role. Findings from a 2002 study of canine athletes appears to support this theory suggesting that excessive activity can lead to muscle strains with resulting inflammation, edema, localized hemorrhage, and eventually fibrosis.¹⁰⁰

Muscle injuries in dogs may be underestimated because of the failure to establish a definitive diagnosis, poor recognition of muscle damage when accompanied by concurrent, more severe trauma, and problems with the classification of muscle trauma.^{99,101,105} Muscle injuries are said to account for only 5% of reported musculoskeletal disease.¹⁰⁷ Lameness is usually more intense in the acute phase and improves with time. When injured muscle

undergoes fibrous contraction, mechanical lameness may remain.^{100,105,108}

Diagnosis

The unique and consistent presentation of gracilis and/or semitendinosus/semimembranosus contracture allows for effective diagnosis by way of a thorough history, observation of gait and physical examination.

Most contractures have a history of acute injury or lameness weeks to months before the onset of contracture. The lameness and initial swelling usually subside with supportive treatment. However, clinical signs of the contracture occur a few weeks later. In most cases there is a distinctive hind limb gait abnormality in affected dogs. The gait is characterized by a shortened stride with a rapid, elastic medial rotation of the paw, internal rotation of the hock with external rotation of the calcaneus and internal rotation of the stifle during the mid to late swing phase of the stride.

On physical examination a firm mass within the gracilis, semitendinosus, or semimembranosus muscle(s) and a taut, firm band along the caudomedial aspect of the thigh are typically palpable. A stretch of the gracilis-semitendinosus muscle complex (hip flexion and stifle extension with limb abduction) may elicit pain and spasm. Stifle extension may be significantly limited in severe cases.

A complete chemistry panel and CBC are usually within normal limits, although creatine phosphokinase (CK) may be elevated. Radiographs of the hindlimb are typically normal; however, an increased radiopacity may be observed between the muscle and the lower tendon junction in some cases. A slightly echogenic muscle may be noted on ultrasound; however, established fibrosis tissue will appear hyperechoic. Magnetic resonance imaging also may be useful for diagnosing this condition in both the acute and chronic phases. Histopathologic examination of the contracted muscle reveals replacement of degenerating myofibers with connective tissue. The findings are consistent with primary myopathy caused by polyphasic muscular damage (degeneration, segmental necrosis, phagocytosis, and fibrosis).

Treatment

Medical management before or in lieu of surgery was attempted in a previous report with no apparent response.⁶ Fifteen dogs had one or more surgical procedures. Although transection, partial excision, or complete resection of the affected muscle resulted in resolution of lameness following surgery, lameness recurred 6 weeks to 5 months (mean, 2.5 months; median, 2 months) following surgery. Adjunctive medical treatment did not prevent recurrence. Variable replacement of the affected muscle with fibrous connective tissue (predominantly along the caudolateral border of the muscle) was evident grossly, and replacement of myofibers

with fibrous connective tissue was confirmed histologically. A definitive etiology could not be established. Myectomy of the entire gracilis muscle still has a poor prognosis because the gait abnormality may return within 3 to 5 months due to semitendinosus involvement.^{6,40}

Aggressive rehabilitation following surgery is necessary to achieve optimal results. Rehabilitation therapy may consist of continuous ultrasound therapy, manual therapy and a home therapeutic exercise program including massage and ROM/stretching. Although rehabilitation therapy may significantly improve the gait abnormality and function of the hindlimb in chronic cases, it may not lead to complete resolution of clinical signs. The sooner the condition is recognized, and treatment is initiated, the better the outcome. Working shepherds with this condition may continue to remain on active duty; however, a continued rehabilitation and home maintenance program is crucial. Several alternative therapies are currently under investigation including shockwave therapy and botox injections. Stem cell therapy is also being considered due to its anti-fibrotic benefits.

Education for all German shepherd and shepherd-related breed owners and handlers is an important component to prevention. Appropriate warm-up, active stretching followed by cool-downs and passive stretching and massage should be performed whenever possible before working and events.

Conclusion

Sporting dogs can have a wide range of sport-related conditions and injuries. A sound knowledge base of structure and biomechanics and of the sport aids in successful rehabilitative management. In addition, rehabilitative needs may change as the patient progresses through therapy and returns to activity. Therefore appropriate and successful rehabilitative management requires constant reevaluation of the patient and modification of the rehabilitation therapy program to address those changing needs.

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34 Neurologic Conditions and Physical Rehabilitation of the Neurologic Patient

William B. Thomas, Natasha Olby, and Lisi Sharon

General Considerations

Typical objectives in rehabilitating the patient with neurologic disease are to minimize pain, reestablish normal neural pathways, prevent secondary complications, and ultimately return the animal to independent function. It is also essential not to worsen neurologic function or pain in patients with spinal instability.

Neurologic conditions are often very painful. Adequate analgesia is important to achieve a successful outcome. Pain-free animals are relaxed and cooperative, recover faster and more completely, and owners are much happier and more compliant with recommendations when their pet is comfortable. Pain originates from multiple sources. Nociceptors respond to a variety of stimuli, including mechanical (e.g., pinching a toe), thermal (heat, cold), and chemical. Activation of superficial mechanical or thermal nociceptors initiates sharp, pricking pain that is precisely localized and ceases when the stimulus is no longer present. Increased intensity of stimulus activates slower conducting pain felt as burning or throbbing that is poorly localized. With tissue injury and inflammation, this type of pain persists even after the initial injury. Patients with neurologic disease can develop neuropathic pain, including spontaneous pain, paresthesia (tingling, prickling), and dysesthesia (paroxysmal shooting, shocklike pain or continuous burning pain). One of the most common causes of neuropathic pain is intervertebral disk herniation that compresses or stretches a nerve root.¹

Analgesic and antiinflammatory drugs are important in managing pain in neurologic patients. Rehabilitation modalities are used to supplement medications and reduce drug requirement. Cryotherapy (see Chapter 18) reduces pain through the gate control theory described by Melzack and Wall that is governed by balance between excitation and inhibition in the dorsal horn, slowed nerve conduction velocity and increased threshold to nerve stimulation. Duration of analgesia lasts at least 30 minutes posttreatment providing an opportunity to perform passive manipulation. Transcutaneous electrical nerve stimulation (see Chapter 20) also works under the same gate control theory (Figure 34-1). Laser therapy (see Chapter 21), and massage

(see Chapter 27) may also reduce pain and enhance relaxation.

Restoring normal function entails a number of exercises, including assisted standing, walking, and swimming. The therapist supports the patient as necessary to give the animal confidence and security when standing and walking. The center of gravity is just behind the scapula, meaning more weight is borne by the thoracic limbs than the pelvic limbs in normal animals. The best way of supporting the animal is to relieve weight from the affected limb(s) and maintain the center of gravity in the correct place by supporting the patient under the sternum and under the pelvis. This minimizes stress on the lumbar vertebrae and allows the cervical vertebrae to move freely. Patients are not supported under the abdomen because this places pressure on abdominal organs and stresses the lumbar vertebrae. Similarly a neck collar is not used to support the patient because this puts pressure on the trachea and cervical vertebrae. A harness that supports the thorax and another to support the pelvis are helpful in larger patients.

When the patient is standing, the therapist helps place the limbs in a normal position. The normal stance is a rectangular configuration in which the feet are at each corner of the rectangle, with each thoracic limb directly under the shoulders and each pelvic limb under the pelvis. The spine should be in a straight line from the head to the tail. With thoracic limb pain, the patient redistributes the weight to the other three limbs, shifting weight and center of gravity caudally, resulting in a stance in the configuration of a triangle. The affected limb is placed laterally, cranially, or caudally and the spine curves toward the unaffected side to maintain equilibrium. This posture overloads the hips and knees. When a thoracic limb is paretic and not painful (e.g., brachial plexus avulsion), the affected limb may be placed caudally and the neck and head may drop ventrally, shifting the center of gravity cranially.

If one pelvic limb is affected, the other pelvic limb is placed medially, the thoracolumbar spine curves to the unaffected side, and the head and neck drop ventrally, shifting the center of gravity cranially. This increases stress on the biceps tendons, elbows, and carpi. When both pelvic limbs are paretic they tend to be placed medially and the thoracic limbs are placed laterally and caudally, with the paws pointing laterally. The head and neck drop ventrally head to shift the center of gravity cranially. This overloads

From Olby N, Halling KB, and Glick TR: Rehabilitation for the Neurologic Patient, Vet Clin North Am Small Anim Pract 35(6):1389-1409, 2005.

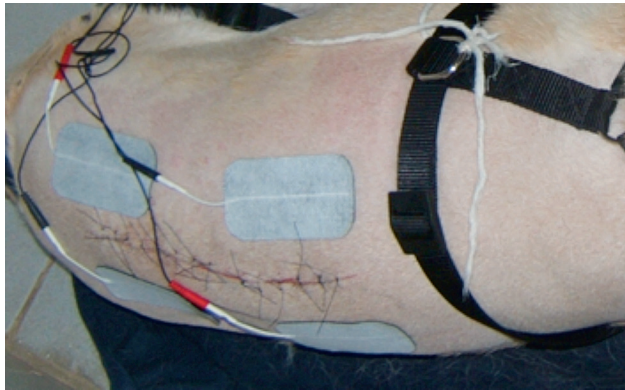


Figure 34-1 Transcutaneous electrical nerve stimulation applied to a patient for pain control.



Figure 34-2 While the patient is eating, the stance may be corrected and gentle weight shifting exercises may be performed.

the shoulders, biceps tendons, elbows, and carpi. The thoracolumbar spine is curved dorsally (kyphosis).

A good time to correct the stance is meal times. Although the patient is occupied with eating, gently correct the stance to a normal position (Figure 34-2). Repeat this as needed to maintain a proper stance.

Neurologic patients are at risk of several complications including shortening of ligaments and tendons, muscle atrophy and contracture, pressure sores, urinary tract infections, and respiratory tract infections that can develop secondary to immobility, recumbency, and altered biomechanical stresses. Modalities such as passive range of motion (PROM) and stretching exercises decrease the risk of contractures. Dry, well-padded bedding and frequently repositioning recumbent patients help prevent pressure sores (Figure 34-3). Bony prominences such as the ischiatic tuberosity, greater trochanter, and xiphoid are inspected frequently to detect early skin lesions. Urine retention is prevented by urethral catheterization or manual bladder expression (see Chapter 16). Helping the patient maintain



Figure 34-3 Dry, well-padded bedding and frequent repositioning of recumbent patients help prevent pressure sores.



Figure 34-4 Patients with loss of sensation or abnormal sensation are at risk of damaging their limbs from dragging or self-mutilation. The affected limb(s) should always be protected to prevent trauma.

a normal, upright posture as much as practical minimizes many of the adverse effects of recumbency. Patients with loss of sensation or abnormal sensation are at risk of damaging their limbs from dragging or self-mutilation (Figure 34-4). In these patients, the affected limb(s) should always be protected to prevent trauma.

Specific neurologic diseases and their management are discussed in the following sections.

Intervertebral Disk Disease

Intervertebral disk (IVD) disease is a syndrome of pain or neurologic deficits, resulting from displacement of part of the disk. Disks are interposed between each vertebral body except the first and second cervical vertebrae and each of the fused sacral vertebrae. The IVD is situated between adjacent vertebral end plates and is composed of an outer fibrous ring, the annulus fibrosus, which surrounds an eccentric amorphous gelatinous center, the nucleus pulposus.²

Intervertebral disk herniation is a general term that refers to displacement of a portion of the disc. Although a normal disk can herniate as a consequence of major trauma, most herniations are secondary to preexisting degeneration of the disk. Disk degeneration is associated with decreased water and proteoglycan concentration. This reduces the ability of the disk to function as a hydraulic cushion and predisposes to displacement of a portion of the disk. There are two types of disk degenerations, chondroid and fibrous metaplasia. Chondroid metaplasia is most common in chondrodystrophic breeds and causes a progressive transformation of the gelatinous nucleus pulposus to hyaline cartilage. This type of degeneration predisposes to disk extrusion where disk material is displaced beyond the outer edge of the annulus (also called Hansen Type I).

Fibrous metaplasia is an age-related degenerative process that occurs in any breed but is more common in nonchondrodystrophic dogs 7 years of age and older. It is characterized by a fibrous collagenization of the nucleus pulposus and degeneration of the annulus fibrosus. Fibrous metaplasia predisposes to disk protrusion, in which disk material is displaced from the disk space but is contained within an intact annulus (also called Hansen Type II). Nonchondrodystrophic breed dogs can also suffer extrusions and less commonly chondrodystrophic breeds suffer protrusions.^{2,3}

The lifetime prevalence for intervertebral disk disease is approximately 3.5% and overall mortality rate attributed to disk disease is about 1%. Miniature dachshunds have the highest risk, with a lifetime prevalence of 20% and other chondrodystrophic breeds are also at increased risk.⁴

Diagnosis is based on clinical features and imaging studies, including radiography, myelography, computed tomography (CT), and magnetic resonance imaging (MRI). Survey radiographs are useful at ruling out other diagnostic possibilities, such as fractures/luxations, vertebral tumors and diskospondylitis, but are usually not accurate enough for surgical planning for intervertebral disk disease. Myelography is more accurate than survey radiographs but carries a risk of complications such as seizures and worsening of neurologic signs. Computed tomography is accurate for acute extrusions in chondrodystrophic breeds and is noninvasive and usually faster than myelography. However, CT is not as sensitive for noncalcified herniations. Magnetic resonance imaging is generally considered the best diagnostic modality and is accurate and noninvasive, but is also expensive.

Cervical Disk Disease

Cervical disk disease commonly affects small, middle-age to older chondrodystrophic breeds, with beagles and dachshunds having the highest risk.^{4,5} The C2-C3 disk space is most commonly affected, with the frequency decreasing at progressively more caudal disk spaces. Cervical disk disease in large-breed dogs is usually associated

with cervical spondylomyelopathy and is discussed separately. Neck pain is the most common and often the only sign of cervical disk disease and is present in approximately 90% of affected dogs.⁵ This is manifested as low head carriage, stiffness or decreased motion of the neck, vocalizing, and spasms of the neck muscles. Some dogs exhibit lameness of one or both thoracic limbs, owing to nerve root or spinal nerve attenuation (root signature). In severe cases, there is ataxia, conscious proprioceptive deficits, weakness, or paralysis of all limbs. Spinal reflexes are usually normal to exaggerated, although reflexes may be weak or absent in the thoracic limbs with caudal cervical disk extrusions and occasionally with more cranial lesions.⁶ Other causes of cervical lesions include trauma, neoplasia, meningitis, diskospondylitis, cervical spondylomyelopathy, and atlantoaxial instability.

Treatment options include surgery and nonsurgical therapy, with the decision based on the duration and severity of clinical signs. Nonsurgical therapy is indicated as the initial treatment in patients with no or mild neurologic deficits. The most important aspect of treatment is strict cage confinement to minimize stress on the damaged disk to allow the disk time to heal. A harness is used instead of a neck collar and leash. A short course of anti-inflammatory doses of corticosteroids or nonsteroidal anti-inflammatory drugs (NSAIDs) is administered if necessary for pain control. Muscle relaxants may be beneficial in some patients with spasm of the cervical muscles. Analgesic drugs without cage confinement are contraindicated because they may lead to increased patient activity and risk further disk extrusion. If the patient does well, cage confinement is continued for at least 2 to 3 weeks. About 50% of dogs with neck pain caused by disk disease recover with nonsurgical therapy. Recurrence of signs occurs in about one third of patients.⁷

Surgery is indicated for patients with substantial neurologic deficits, neck pain unresponsive to appropriate nonsurgical therapy, or recurrent bouts of neck pain. Removal of extruded disk material by a ventral slot to decompress the spinal cord and nerve roots is the procedure of choice in most patients. Dorsal hemilaminectomy may be necessary for lateral or far-lateral extrusions. In chondrodystrophic dogs, prophylactic fenestration of the other disk spaces at the time of ventral slot is indicated to decrease the risk of extrusion of another disk. Fenestration alone of the affected disk is not indicated because it does not allow removal of extruded disk material.⁸ In dogs with neck pain and mild neurologic deficits, the success rate with surgery is about 95%. Nonambulatory dogs have a good outcome in 62% to 100% cases, with small dogs (≤ 15 kg) having a better prognosis than larger dogs.^{5,8,9} Preoperative, intraoperative, and postoperative management included drugs described for medical management. Postoperative physical rehabilitation is an integral part of the overall care of the patient.



Figure 34-5 Whirlpool with hydraulic hoist and sling.

Physical Rehabilitation for Cervical Disk Disease

The most severe cervical injuries cause tetraplegia with a compromise of respiratory function. It is imperative that respiratory function is evaluated in any tetraplegic animal and that respiratory complications (such as pneumonia or hypoventilation) are identified and understood before any exercises that may exacerbate the problem.

Physical rehabilitation as part of medical management includes massage, heat therapy, electrical stimulation (ES), therapeutic ultrasound (US), therapeutic laser for muscle spasms and pain relief. Whirlpool therapy may also be cautiously used in patients carefully supported in a sling as long as there is no struggling (Figure 34-5).

Postoperative therapy is staged and begins immediately or within 48 hours of surgery to relieve pain and muscle spasms. Modalities such as cryotherapy, ES, carefully administered US, laser and massage may be used based on the clinical needs of the patient. In ambulatory patients, neck leashes should not be used, and leashes should be attached to a harness or across the chest and under one thoracic limb.

Methods of actively eliciting cervical motion begin in the first few days after comfort is achieved and there is no pain on movement of the neck. Active range of motion can be achieved using toys or treats to motivate the patient to turn the head and neck in all directions. Begin with smaller movements and progress to the point that the patient is able to take a toy or treat from near the rear limbs for lateral excursion, between the forelimbs for cervical flexion, and high above the head for extension. Pain-free PROM, stretching, and massage are used to help treat muscle spasms.

Resistive exercises are used in tetraparetic, paraparetic, and hemiparetic patients. Pushing on the bottom of the foot of an extended extremity elicits a flexor withdrawal reflex causing flexion of the limb that stimulates muscles, tendons, ligaments, joints, and bones of the limb. While maintaining



Figure 34-6 Life preserver used for flotation.

limb extension and force on the foot, move the limb slightly in all directions. As the reflex causes the limb to retract, active, resistance can be achieved by the therapist holding the foot, creating a gentle “tug-of-war” in which the patient is pulling more forcefully to withdraw the limb from the therapist’s grip. This exercise should be performed for three to five repetitions per limb, three to four times per day.

Assistance should be provided to patients that are able to stand and support weight to attain correct limb placement. Careful use of a sling and hoist may be useful in larger nonambulatory patients. Necessary support through the chest is provided as needed to invoke as much independent weight bearing as possible. Patients are assisted in slow ambulation by placing and moving any limbs with motor deficits. Assisted walking (tail, sling), proprioception stimulation, and limb placement exercises facilitate neuromuscular re-education and proprioceptive functioning, as well as muscle strengthening.

Spinal walking is a phenomenon that occurs in cats following surgical transection of the spinal cord and occurs in some dogs after an injury that causes severe spinal cord injury or transection. While these dogs do not regain normal motor function, spinal walking may be a functional mode of ambulation for short distances.

For strengthening, underwater treadmill walking or swimming may be initiated after the incision site is healed. The time of suture removal is generally a safe time to start hydrotherapy. Some therapists start some form of aquatic therapy earlier, with caution to avoid submersion of the incision, especially an incision in the ventral cervical region. In any case, the incision should be sealed, with apposed edges, no discharge or drainage, and no separation of the incision with gentle manipulation of the edges. In small dogs, a whirlpool may be used for aquatic therapy. The duration of aquatic therapy depends on the fitness, endurance, and ability of the patient. Assistance or life preservers are provided as required (Figure 34-6). If the patient does not actively move its limbs through an

adequate range of motion, assistance is given either by passively moving the limb(s) or by providing resistance to motion to stimulate the patient to engage the limb against the applied resistance.

Patients can be placed on exercise rolls to strengthen proprioception and joint stability. They may be further challenged with mild bouncing and slight movements of the ball to create an unstable surface (Figure 34-7). Requiring patients to walk on foam, bubble wrap, or an inflatable object or challenging them with a rocker board (Figure 34-8) will also help them to develop stability and proprioception. To assist in regaining kinesthetic awareness, patients can be challenged to step over objects of varying shapes and sizes (Figure 34-9). Small objects are initially used; the height and width are increased as the patient progresses. Strengthening the thoracic limbs can be achieved by ambulating on land, a dry treadmill (Figure 34-10), and deep water and underwater treadmill hydrotherapy. Additional strengthening exercises include walking down inclines with variable pitch, wheel

barrowing (Figure 34-11), and increased weight bearing with an exercise roll beneath the pelvic limbs (Figure 34-12). Analogous exercises may be used to strengthen the rear limb muscles, such as walking up inclines, dancing, and weight bearing with the forelimbs on an exercise roll.

Therapeutic modalities are provided for a minimum of 3 weeks. Owners are instructed further in performing strengthening exercises after discharge. Continued rehabilitation modalities are often prescribed after discharge on a recommended weekly schedule.

Thoracolumbar Disk Disease

Thoracolumbar herniations are most common from T12-T13 to L1-L2, although other areas may also be affected. dachshunds, cocker spaniels, and beagles and other chondrodystrophic breeds are at increased risk.⁴ Type I extrusions are most common in these breeds. Type II protrusions are more common in large, nonchondrodystrophic dogs, but large dogs can also suffer acute, Type I extrusions.¹⁰

Onset of signs can be sudden or gradual and depend on the force, quantity, and location of the disk material and the duration of disk herniation. About 10% of affected dogs have back pain with no neurologic deficits. They are often reluctant to run, jump, or climb stairs and may have kyphosis. Neurologic deficits range from mild ataxia of the pelvic limbs to complete paraplegia with urinary and fecal incontinence and absent deep-pain perception. The neurologic deficits may be symmetric or worse on one side. Spinal reflexes are usually normal to exaggerated, but are weak to absent if the extrusion occurs caudal to L2-L3. In the most severe cases, progressive ascending-descending myelomalacia may result. In this syndrome, the spinal cord is damaged progressively cranial and caudal from the initial site. This is recognized in patients with an initial thoracolumbar lesion as a loss of pelvic limb reflexes, loss of abdominal muscle tone, cranial migration of the level of anesthesia and the cutaneous tunic reflex, thoracic limb paresis, and ultimately respiratory arrest. This is a grave condition that the therapist should be aware of if the neurologic status deteriorates.



Figure 34-7 Therapy roll used for improving conscious proprioception.

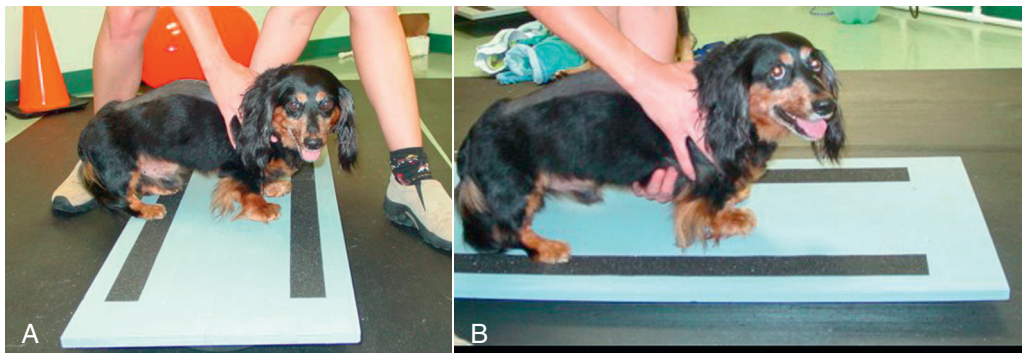


Figure 34-8 Rocker boards for rehabilitation. **A**, Side-to-side motion. **B**, Forward and backward motion.



Figure 34-9 Various-sized rails used as obstacles.



Figure 34-10 Land treadmill.



Figure 34-11 Wheelbarrow exercises.

Other diagnostic considerations for thoracolumbar spinal lesions include trauma, fibrocartilaginous embolism, degenerative myelopathy, discospondylitis, degenerative lumbosacral stenosis, neoplasia, meningitis, and orthopedic conditions such as severe hip dysplasia or bilateral cranial cruciate ligament insufficiency.



Figure 34-12 Facilitating patient standing using a therapy roll.

About 85% of dogs with back pain and no neurologic deficits will improve with nonsurgical therapy, as described for cervical disk disease. Surgery is indicated in patients with substantial neurologic deficits, or back pain refractory to nonsurgical therapy. Patients with a sudden onset of severe neurologic deficits need urgent surgery. Removal of extruded disk material via hemilaminectomy is the procedure of choice in most cases. Fenestration of the affected intervertebral disk space is routinely performed. Prophylactic fenestration of other thoracolumbar disk spaces reduces recurrence in chondrodystrophic breeds.¹¹ The prognosis depends in large part on the severity and duration of the neurologic deficits. The recovery rate for nonambulatory dogs with acute extrusions and intact deep pain perception is 85% to 95%, with most patients regaining ambulation in 1 to 4 weeks. For dogs with loss of deep pain perception, the recovery rate is about 50% if surgery is performed within 24 hours of the loss of deep pain. This decreases further if surgery is delayed for longer than 24 hours. The recovery rate for large breed dogs with chronic protrusions is much lower, ranging from 22% to 52%.⁴

Physical Rehabilitation for Thoracolumbar Disk Disease

Although physical rehabilitation has been used with pharmacologic intervention in the conservative management of acute and recurrent thoracolumbar disk disease with some success, it is important to perform minimal intervention and avoid motions that might risk further herniation of disk material for one month; the major application of therapeutic modalities is in the postoperative period. The initiation of therapy and protocols used are similar to those used for cervical disk patients, but the focus is on the pelvic limbs. Patients with thoracolumbar disk disease are more likely to have proprioceptive and motor deficits, requiring greater attention to balancing and strengthening activities. Additional nursing care is required such as